

Document Title

**Cloud-Based Custom Architects  
Management System**

Subtitle

**Nirman Architects Management System**

Prepared For

**Hirak bhai**

Prepared By

**Nexalliance - Your Trusted IT Partner**

# 1. Project Overview

## Introduction

The **Nirman Architects Project Management & Workforce Management System** is a centralized cloud-based platform designed to digitize and streamline the complete operational workflow of the organization.

The system will replace multiple disconnected tools with a single integrated platform that manages projects, employees, clients, drawings, attendance, productivity, communication, documentation, and reporting.

The solution will be available as:

- Web Application (Admin & Staff)
- Android Application
- iOS Application
- Customer Portal (Web & Mobile)

The platform will provide real-time visibility into every project, employee performance, drawing status, task progress, approvals, and site activities while ensuring accountability and transparency throughout the organization.

## 2. Current Software Being Used

Currently, Nirman Architects uses the following software:

### **Desklog.io**

Used for:

- Employee Attendance
- Time Tracking
- Random Screenshots
- Productivity Monitoring
- PC Activity Tracking

### **Archizy**

Used for:

- Project Management
- Task Management
- Drawing Management

The proposed solution aims to consolidate these functionalities into one unified platform while adding several advanced features.

### 3. Vision of the System

To build a modern digital ecosystem that enables seamless collaboration between management, architects, designers, engineers, site teams, clients, and stakeholders by automating project workflows, employee management, drawing approvals, and communication.

### 4. Objectives

The primary objectives of the system include:

- Centralized Project Management
- Employee Productivity Monitoring
- Smart Attendance Management
- Drawing Version Control
- Customer Collaboration
- Site Monitoring
- Task Automation
- Project Timeline Tracking
- Approval Workflow Management
- Performance Analytics
- Secure Document Management
- Mobile Accessibility
- Real-time Notifications
- Business Intelligence Dashboard

### 5. User Roles

The system will support role-based access.

#### Super Admin

- Complete System Control
- HR Management
- Employee Management
- Reports
- Settings
- Permissions
- Analytics

## **HR**

- Employee Attendance
- Leave Management
- Payroll Data
- Employee Monitoring
- Working Hours
- Performance Reports

## **Project Manager**

- Create Projects
- Assign Teams
- Monitor Progress
- Review/Manage Drawings
- Approvals
- Client Communication
- Reports

## **Architect / Designer**

- Receive Tasks
- Upload Drawings
- Submit Drawings
- Update Status
- Chat
- Time Tracking

## **Site Engineer**

- Site Updates
- Progress Photos
- Site Attendance
- Issue Reporting
- Client Updates

## **Employee**

- Attendance
- Daily Tasks
- Drawings
- Chat
- Documents
- Notifications

## Customer

- View Project Progress
- View Drawings
- Download Documents
- Chat
- Approval
- Raise Queries

## 6. Dashboard

The Dashboard should provide a complete real-time overview.

### Admin Dashboard

- Total Projects
- Active Projects
- Completed Projects
- Delayed Projects
- Pending Approvals
- Today's Attendance
- Employee Productivity
- Online Employees
- Employees Working on Site
- Employees Working from Office
- Recent Activities
- Notifications
- Upcoming Deadlines
- Project Progress
- Revenue Dashboard
- Task Summary
- Drawing Status
- Project Health Score

## 7. Project Management Module

The Project Management module will manage the complete lifecycle of every project.

### Features

#### Project Creation

- Project Details

- Client Information
- Address
- Timeline
- Budget
- Team Assignment
- Priority
- Project Category

### **Project Timeline**

- Start Date
- Estimated Completion
- Actual Completion
- Delays
- Milestones
- Progress Percentage

### **Team Assignment**

- Multiple Employees
- Role Assignment
- Department Assignment
- Responsibility Matrix

### **Project Status**

- New
- Planning
- In Progress
- On Hold
- Approval Pending
- Site Work
- Completed
- Archived

### **Project Progress Tracking**

- Overall Progress
- Department-wise Progress
- Employee-wise Progress
- Drawing-wise Progress
- Task-wise Progress

## **8. Task Management System**

The task management module will provide detailed work allocation and monitoring.

## Features

### Task Creation

- Task Name
- Description
- Priority
- Department
- Assigned Employee
- Estimated Time
- Deadline
- Attachments

### Task Workflow

Pending



Accepted



In Progress



Review



Approved



Completed



### Task Features

- Task Acceptance
- Task Rejection
- Task Reassignment
- Comments
- Attachments
- Checklists
- Dependencies
- Time Tracking
- Due Date Reminder

## Time Analysis

After task acceptance, the system should automatically calculate:

- Actual Start Time
- Completion Time
- Total Working Time
- Delay
- Idle Time
- Productivity Score

## Overdue Task Management

The system should automatically detect:

- Overdue Tasks
- Delayed Projects
- Pending Approvals
- Missed Deadlines

Automatic reminders should be sent to:

- Employee
- Project Manager
- Admin

# 9. Drawing Management System

This is one of the core modules, which contains **Project wise** and **Person wise** (Customer name).

## Drawing Categories

- Concept Drawings
- Working Drawings
- Process DWG
- GFC Drawings
- Site
- Interior Drawings

## Drawing Version Control

Every uploaded drawing should automatically generate:

- Version Number
- Upload Date
- Uploaded By



- Revision History
- Change Logs

No drawing should ever be permanently replaced.

Previous versions should always remain accessible.

## Drawing Approval Workflow

Designer Uploads

↓

Project Manager Review/Manage

↓

Admin Review/Manage

↓

Approved

↓

Visible to Staff & Client (as permitted)

## Process DWG Editing

Only **Admin** (or authorized users) can edit **Process DWG** files.

Employees can:

- View
- Download
- Comment
- Compare Versions

Editing rights remain restricted.

## GFC Drawing Management

- Final Approved Drawings
- Locked Version
- Client Accessible
- Downloadable
- Revision Tracking

## 10. JPEG / 3D Drawing Review

3D renders will be uploaded in JPEG format.

Features:

- Zoom
- Full Screen
- Image Annotation
- Marking Tools
- Highlight Area
- Notes
- Approval
- Comments
- Version History

Clients and internal teams can review the same image collaboratively.

## 11. Project Chat System

Each project will have its own communication workspace.

Members included:

- Admin
- Project Manager
- Assigned Staff
- Customer

Features:

- Text Chat
- Mention Users
- Reply Message

All communication remains project-specific.

## 12. Employee Attendance Management

Attendance will be fully automated.

### Office Attendance

When the employee turns ON the company laptop:

- Auto Check-in
- Attendance Started

When the laptop is turned OFF:

- Auto Check-out

Working hours calculated automatically.

## **Site Attendance**

Employees working at client locations must:

- Check-in using Mobile
- GPS Verification
- Geo-fencing
- Timestamp
- Selfie Verification (Optional)
- Check-out from Site

## **Attendance Reports**

- Daily
- Weekly
- Monthly
- Employee Wise
- Department Wise

# **13. Employee Productivity Monitoring**

The system should continuously monitor employee productivity.

## **PC Tracking**

- Active Time
- Idle Time
- Application Usage
- Website Usage
- Keyboard Activity
- Mouse Activity
- Productive Hours
- Non-Productive Hours
- Focus Time
- Break Time

## Random Screenshots

The desktop application should automatically capture screenshots at random intervals during working hours.

Settings:

- Configurable Interval
- Blur Sensitive Applications (Optional)
- Admin Controls

## Productivity Analysis

Each employee should have:

- Daily Productivity
- Weekly Productivity
- Monthly Productivity
- Utilization Percentage
- Focus Time
- Idle Time
- Attendance
- Screenshot History
- Task Completion Ratio

## 14. HR Management

Features:

- Employee Master
- Departments
- Designations
- Joining Date
- Leave Management
- Holiday Calendar
- Attendance
- Shift Management
- Payroll Data
- Performance Reviews
- Employee Documents

## 15. Project Analysis

Each project should provide detailed analytics.

## Project Dashboard

- Progress
- Completion %
- Delays
- Pending Tasks
- Employee Performance
- Drawing Status
- Budget Tracking
- Timeline Tracking

## Employee-wise Analysis

- Assigned Tasks
- Completed Tasks
- Delayed Tasks
- Average Completion Time
- Productivity
- Attendance
- Working Hours

## Task-wise Analysis

- Estimated Hours
- Actual Hours
- Delay
- Employee Responsible
- Status
- Priority

# 16. Reports

The system should generate comprehensive reports.

Reports include:

- Attendance Report
- Productivity Report
- Project Report
- Employee Report
- Drawing Report
- Site Report
- Daily Progress Report
- Monthly Progress Report
- Customer Report

- Task Report
- Approval Report

Export formats:

- PDF
- Excel
- CSV

## 17. Document Management

Every project should maintain its own document repository.

Supported files:

- PDF
- DWG
- JPEG
- PNG
- DOCX
- XLSX
- ZIP

Features:

- Folder Structure
- Date-wise Labels
- Version Control
- Download History
- Upload History
- Search Documents
- Preview Files
- Access Permissions

## 18. Notification System

Notifications through:

- Mobile Push Notifications
- Email
- In-App Notifications
- Whatsapp Notifications (**API required from Client side**)

Alerts include:

- New Task

- Approval Required
- Drawing Uploaded
- Drawing Approved
- Task Overdue
- Project Delay
- Employee Check-in
- Client Comments
- Deadline Reminder

## 19. Customer Portal

Customers can:

- View Project Dashboard
- Track Progress
- View Timeline
- View Drawings
- Download PDFs
- Download Photos
- View 3D JPEGs
- Approve Drawings
- Chat with Team
- Raise Queries
- Receive Notifications
- Access Project History

Available on:

- Android
- iOS
- Web



## 20. Mobile Applications

Dedicated Android and iOS applications will support:

- Attendance
- Geo-location Check-in
- Task Management
- Project Updates
- Drawing Viewer
- PDF Viewer
- Chat
- Notifications
- Site Photo Upload

- Approval Workflow
- Offline Data Sync (for low-connectivity sites)
- Smooth viewing of large drawing files with pinch-to-zoom and optimized rendering

## 21. Integration Requirements

The system should support integration with:

- Email Services
- SMS Gateway
- WhatsApp Notifications
- Calendar Integration (Optional)
- Digital Signature (Future)
- Accounting Software (Future)

## 22. Future Enhancements

The platform should be scalable to include the following advanced capabilities:

- AI-based Project Delay Prediction
- AI-powered Resource Allocation
- Auto Task Scheduling
- AI Drawing Comparison & Change Detection
- OCR for Scanned Drawings and Documents
- Voice-to-Task Creation
- BIM (Building Information Modeling) Integration
- CAD Viewer Integration for DWG Files
- Drone Progress Monitoring
- Site Inspection Checklists
- Material & Inventory Management
- Vendor & Contractor Management
- Purchase Order & Procurement Workflow
- BOQ (Bill of Quantities) Management
- Cost Estimation & Budget Variance Tracking
- Invoicing & Billing Module
- Digital Approval with E-Signatures
- QR Code-Based Document Retrieval
- Customer Feedback & Satisfaction Dashboard
- KPI Dashboard for Departments
- Employee Performance Scoring with AI
- Automated Daily Progress Reports (DPR)
- Meeting Minutes (MoM) Management
- Risk & Issue Register
- Asset & Equipment Tracking
- Visitor Management for Office/Sites



- Multi-Branch Management
- Multi-Company Support
- API Integration with Third-Party Software
- Business Intelligence (BI) Dashboard
- Predictive Analytics and Executive Reporting

## 23. Expected Business Benefits

Implementing this integrated platform will provide:

- A single unified system replacing multiple software solutions.
- End-to-end visibility across projects, drawings, employees, and client communication.
- Improved accountability through automated attendance, productivity monitoring, and audit trails.
- Faster project delivery with structured workflows, approval tracking, and real-time collaboration.
- Reduced operational delays through automated reminders, overdue alerts, and project analytics.
- Better client experience with transparent progress tracking and centralized communication.
- Enhanced decision-making using real-time dashboards, KPI reports, and performance insights.
- Scalable architecture capable of supporting future business growth, new branches, and additional services.

## 24. Conclusion

The proposed Nirman Architects Project Management & Workforce Management System is designed to serve as a complete digital ecosystem for managing architectural projects, workforce productivity, client collaboration, and organizational operations from a single centralized platform.

By integrating project management, drawing workflows, employee attendance, productivity monitoring, task tracking, approval management, customer communication, and analytical reporting into one system, Nirman Architects will significantly reduce manual processes, improve operational transparency, and enhance project execution efficiency.

This solution will empower Nirman Architects to operate more efficiently, make data-driven decisions, improve team collaboration, and deliver projects with greater accuracy, transparency, and professionalism.

## 25. Contact Information

**Nexalliance - Your Trusted IT Partner**

Email: [info@nexallianceit.com](mailto:info@nexallianceit.com)

Phone: +91 63511 78511

Website: <https://nexallianceit.com/>

